BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

CHEMISTRY

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Chemistry (Objective) Group - I

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8483

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. During the manufacturing process of cement the temperature of decomposition zone goes up to:

(A) 600°C
(B) 900°C
(C) 1000°C
(D) 1200°C

Q2. Carboxylic acids on reduction with HI and red phosphorus gives:

(A) Alkanes
(B) Alcohols
(C) Aldehydes
(D) Ketones

Q3. Which acid is used in the manufacture of synthetic fiber?

(A) Formic acid
(B) Oxalic acid
(C) Carbonic acid
(D) Acetic acid

Q4. The compound used in the processing of anti-polio vaccine is:

(A) Acetaldehyde
(B) Formaldehyde
(C) Acetone
(D) Ethyl bromide

Q5. Formalin is _____ solution of Formaldehyde in water.

(A) 10%
(B) 20%
(C) 40%
(D) 60%

Q6. Which compound will have maximum repulsion with H_2O?

(A) C_6H_6
(B) C_2H_5OH
(C) $CH_3CH_2CH_2OH$
(D) $CH_3–O–CH_3$

Q7. Which is not a nucleophile?

(A) H_2O
(B) H_2S
(C) BF_3
(D) NH_3

Q8. The electrophile in aromatic sulphonation is:

(A) H_2SO_4
(B) HSO_4^-
(C) SO_3
(D) SO_3^+

Q9. Formula of chloroform is:

(A) CH_3Cl
(B) CCl_4
(C) CH_2Cl_2
(D) $CHCl_3$

Q10. A double bond consists of:

(A) Two sigma bonds
(B) One sigma and one pi bond
(C) One sigma and two pi bonds
(D) Two pi bonds

Q11. The colour of transition metal complexes is due to:

(A) d-d transition of electrons
(B) Paramagnetic nature of transition elements
(C) Ionization
(D) Loss of S-electrons

Q12. The anhydride of $HClO_4$ is:

(A) ClO_3
(B) ClO_2
(C) Cl_2O_5
(D) Cl_2O_7

Q13. Which halogen is a solid at room temperature and pressure?

(A) F_2
(B) Cl_2
(C) Br_2
(D) I_2

Q14. Among group VA elements, the most electronegative element is:

(A) Sb
(B) N
(C) P
(D) As

Q15. Tincal is a mineral of:

(A) Al
(B) B
(C) Si
(D) C

Q16. Chile Saltpeter has the chemical formula:

(A) $NaNO_3$
(B) KNO_3
(C) $Na_2B_4O_7$
(D) $Na_2CO_3 \cdot H_2O$

Q17. Mark the correct statement:

(A) Metallic character increases down the group
(B) Metallic character increases from left to right along a period
(C) Metallic character remains the same from left to right along a period
(D) Metallic character remains the same down the group

Chemistry (Subjective) Group - I

Time Allowed: 02:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 8483

Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT parts.

(i) Why anionic radius is greater than parent atom?

(ii) Diamond is a non-conductor while graphite is a good conductor. Give reason.

(iii) Complete and balance the equations: (a) $LiNO_3$ --heat--> (b) $NaNO_3$ --heat-->

(iv) Describe two problems during manufacturing of NaOH by diaphragm cell.

(v) Convert Boric acid into tetra boric acid.

(vi) Write the reaction of H_3BO_3 with (a) NaOH (b) Na_2CO_3

(vii) Write any two uses of boric acid.

(viii) Write two methods for preparation of nitrogen oxide (NO).

(ix) Write any two reactions of H_2SO_4 as an oxidizing agent.

(x) How diammonium phosphate is prepared?

(xi) Define cement.

(xii) Which types of raw material is used in cement? Give their names.

Q3. (Marks 16) Write short answers to any EIGHT parts.

(i) Write equations for the reactions of chlorine with hot and cold NaOH.

(ii) Give four uses of bleaching powder.

(iii) Arrange the oxy acids of halogen in increasing order of their acidic strength.

(iv) What is sacrificial corrosion?

(v) What are interstitial compounds?

(vi) Write mechanism for nitration of benzene.

(vii) Convert benzene into (a) Hexachlorocyclohexane (b) Benzene sulphonic acid.

(viii) What is Tollen's test?

(ix) Write general mechanism for the acid catalysed nucleophilic addition reactions of carbonyl compounds.

(x) Write four uses of acetic acid.

(xi) Convert acetic acid into (a) Ethane (b) Ethyl alcohol.

(xii) Write structural formulae of (a) Malonic acid (b) Phthalic acid.

Q4. (Marks 12) Write short answers to any SIX parts.

(i) Define heterocyclic compounds and give two examples with names.

(ii) What is metamerism? Give one example.

(iii) Write the structural formulas for these compounds. (a) 3-n-propyl-1, 4-pentadiene (b) But-1-en-3-yne

(iv) How will you convert? (a) Ethene into ethyl alcohol (b) Ethene into ethyne.

(v) Define Markownikov's rule and give one example.

(vi) Define allyl halide, which is the best method of preparing allyl halide.

(vii) Give IUPAC names of following compounds: (a) $(C_2H_5)_2CH - CH_2 - CH(Cl) - CH_3$ (b) $(CH_3)_2CH - CH_2 - CH(C_2H_5)CH_2Cl$

(viii) How phenol is prepared from sodium salt of benzene sulphonic acid?

(ix) Give uses of ethanol. Only four.

> Note: SECTION - II Attempt any THREE questions. Each question carries 08 marks.

Q5(a). (Marks 04) Describe variation of melting point and boiling point in periods and groups of modern periodic table.

Q5(b). (Marks 04) Describe peculiar behaviour of Be.

Q6(a). (Marks 04) Write preparation and two reactions of HNO_2.

Q6(b). (Marks 04) Write a note on these properties of transition elements: (i) Binding energies (ii) Oxidation state

Q7(a). (Marks 04) Explain geometrical isomerism with suitable examples.

Q7(b). (Marks 04) What is Cannizzaro's reaction? Explain with mechanism.

Q8(a). (Marks 04) Describe any four methods for the preparation of alkenes.

Q8(b). (Marks 04) What is B-Elimination reaction? Explain E_2 reaction in detail.

Q9(a). (Marks 04) What are Friedel and Craft's reactions? Give one example in each case with mechanism.

Q9(b). (Marks 04) How will you obtain pure ethanol by fermentation of starch.

GROUP – II

Chemistry (Objective) Group - II

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8488

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. Preparation of vegetable ghee involves:

(A) Halogenation
(B) Hydrogenation
(C) Hydroxylation
(D) Dehydrogenation

Q2. Which is meta-directing group?

(A) –OH
(B) $–NH_2$
(C) $–NO_2$
(D) –Cl

Q3. For which mechanism, first step involved is same?

(A) E_1 and E_2
(B) E_2 and SN_2
(C) E_1 and SN_1
(D) SN_1 and E_2

Q4. Ethanol can be converted into ethanoic acid by:

(A) Hydrogenation
(B) Oxidation
(C) Hydration
(D) Halogenation

Q5. The homologous series of both aldehydes and ketones have the general formula:

(A) $C_nH_{2n}O_2$ (B) C_nH_{2n+2}

(C) $C_nH_{2n}O$ (D) C_nH_{2n-2}

Q6. Formalin is _____ solution of Formaldehyde in water.

(A) 10% (B) 20%

(C) 40% (D) 60%

Q7. Alkane nitrile can be converted into carboxylic acids by:

(A) Hydration (B) Acidic hydrolysis

(C) Hydrogenation (D) Oxidation

Q8. Which reagent is used to covert a carboxylic acid to an alcohol?

(A) H_2/Ni (B) H_2/Pt

(C) $NaBH_4$ (D) $LiAlH_4$

Q9. All the nitrogen fertilizers except _____ make the soil acidic.

(A) Calcium nitrate (B) Ammonium nitrate

(C) Potassium nitrate (D) All these

Q10. Ionization energy of calcium is lower than _____ element.

(A) Strontium (B) Magnesium

(C) Barium (D) Sodium

Q11. Which compound is added in Down's cell to lower the melting point of sodium chloride?

(A) $CaSO_4$ (B) $CaCl_2$

(C) $Ca(NO_3)_2$ (D) Na_2CO_3

Q12. Boric acid cannot be used:

(A) As antiseptic in medicine (B) For washing eyes

(C) In soda bottles (D) For enamels and glazing

Q13. Nitric acid does not react with all metals given, except:

(A) Gold (B) Platinum

(C) Magnesium (D) Iridium

Q14. Chlorine heptoxide (Cl_2O_7) reacts with water to form:

(A) Hypochlorous acid (B) Chloric acid

(C) Perchloric acid (D) Chlorine and oxygen

Q15. Cl cannot oxidize:

(A) F^- (B) Br^-

(C) I^- (D) Na-metal

Q16. The colour of $[Ti(H_2O)_6]^{3+}$ ion is:

(A) Red (B) Yellow

(C) Violet (D) Green

Q17. Fredrick Wholer synthesized urea by heating:

(A) NH_4Cl (B) $(NH_4)_2CO_3$

(C) NH_4CNO (D) NH_3

Chemistry (Subjective) Group - II

Time Allowed: 02:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 8488

Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT parts.

(i) The oxidation states vary in a period but remain almost constant in a group. Why?

(ii) The hydration energies of the ions are in the following order $Al^{3+} > Mg^{2+} > Na^+$. Justify.

(iii) Write the names and chemical formulas of important minerals of sodium.

(iv) What happens when (a) Lithium carbonate is heated (b) Lithium hydroxide is heated to red hot?

(v) How borax can be converted into orthoboric acid?

(vi) Why nitric acid is frequently transported in Aluminium containers?

(vii) Write the names and chemical formulas of four important boric acids.

(viii) How does nitrogen differ from other elements of its group?

(ix) Write the equation for the reaction between conc·H_2SO_4 and copper and explain what type of reaction is it?

(x) Why nitrogen is necessary for plants? Give names of two nitrogen fertilizers.

(xi) What do you mean by setting of cement?

(xii) Write any four points of essential qualities of a good fertilizer.

Q3. (Marks 16) Write short answers to any EIGHT parts.

(i) Give reactions of bleaching powder with ammonia and carbon dioxide.

(ii) Write reaction for the preparation of bleaching powder and names of the methods used to prepare it.

(iii) Write names of these compounds. (a) $NaClO_3$ (b) HIO_3

(iv) Why does damaged tin plated iron get rusted quickly?

(v) What are substitutional alloys? Give examples.

(vi) What arguments were given by Kekule to confirm the regular hexagonal structure for benzene?

(vii) How would you prepare benzene from acetylene and toluene from n-heptane?

(viii) What is Benedict's solution test?

(ix) Write the names of those weak oxidizing agents, which can oxidize aldehydes but not the ketones?

(x) Write molecular formulas of palmitic acid and stearic acid.

(xi) How would you prepare acetic acid from ethanol and a suitable alkane nitrile?

(xii) Give two reactions of carboxylic acids in which OH group of the acids are involved.

Q4. (Marks 12) Write short answers to any SIX parts.

(i) Define functional group. Give any two examples.

(ii) What is tautomerism? Give example.

(iii) What is hydrogenolysis? Give example.

(iv) Convert CH_4 into formaldehyde by catalytic oxidation.

(v) What is mustard gas? How it is prepared?

(vi) Define nucleophile and electrophile.

(vii) Complete the reactions: (a) $C_2H_5Br + NH_3 \rightarrow$ (b) $C_2H_5Br + CH_3COONa \rightarrow$

(viii) Write the structural formulae of lactic acid and tartaric acid.

(ix) Distinguish between methanol and ethanol by one test.

> Note: SECTION - II Attempt any THREE questions. Each question carries 08 marks.

Q5(a). (Marks 04) Write similarities and differences of hydrogen with IV-A group elements.

Q5(b). (Marks 04) Describe diaphragm cell method for preparation of NaOH.

Q6(a). (Marks 04) Write equations for the reactions of conc·HNO_3 with: (i) Zn (ii) Cu (iii) Sn (iv) HI

Q6(b). (Marks 04) Describe electrochemical theory to explain corrosion.

Q7(a). (Marks 04) Define atomic orbital hybridization. Explain sp^2-hybridization.

Q7(b). (Marks 04) What is aldol condensation reaction? Give an example and mechanism.

Q8(a). (Marks 04) Explain Markownikov's rule with mechanism and two examples.

Q8(b). (Marks 04) Explain S_N2 reactions with example and characteristics.

Q9(a). (Marks 04) How straight chain structures for the benzene is ruled out.

Q9(b). (Marks 04) Give the preparation of methyl alcohol on large scale. How it may be distinguished from ethyl alcohol.